

Temperature & Humidity Sensor with RS485 (Modbus RTU)

1. Product Overview

WIN-SN-TNH-M-D is a versatile and reliable solution for anyone needing **precise temperature and humidity** monitoring with robust communication capabilities. It measures temperature and humidity in various environments, from industrial facilities to greenhouses and smart homes. This board stands out because it can communicate its readings through an isolated **RS485** interface. This type of communication is robust, making it ideal for long-distance data transmission and resistant to electrical interference, ensuring accurate and reliable measurements.

One of the key advantages of the WIN-SN-TNH-M-D is its flexibility in sensors compatibility. It supports a variety of sensors those are listed in Table 1 each suited for different range of temperature and humidity.



This manual provides instructions for installation and use of the sensor board to measure temperature and humidity in the environment. The sensor board offers data output in Modbus RTU (RS485) and can be powered by DC power supply.

	Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
☐	AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
☐	AHT20	-40°C ~ 85°C.	0~80% RH.	-40°C ~ 85°C.	0.3°C, 2.0 %RH
☐	SHT21	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.3°C, 2.0 %RH
☐	SHT25	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.2°C, 1.8 %RH

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*40 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

5. Modbus RS485 Data Storage register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40001	0x03	RS485	16-bit int
2	Humidity	40002	0x03	RS485	16-bit int
1	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
2	Enter New Baud Rate	40012	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
4	Enter New Slave ID	40014	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
6	Enter New Parity	40016	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

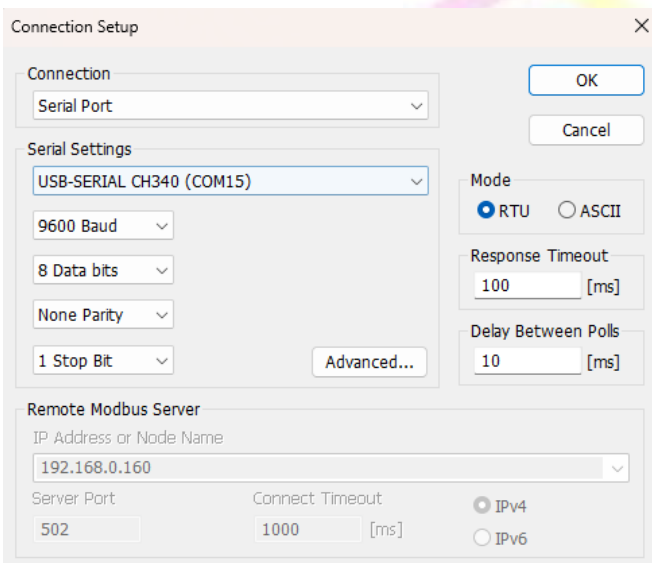
6. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.



Connection Setup

Connection: Serial Port

Serial Settings: USB-SERIAL CH340 (COM15)

Mode: ☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

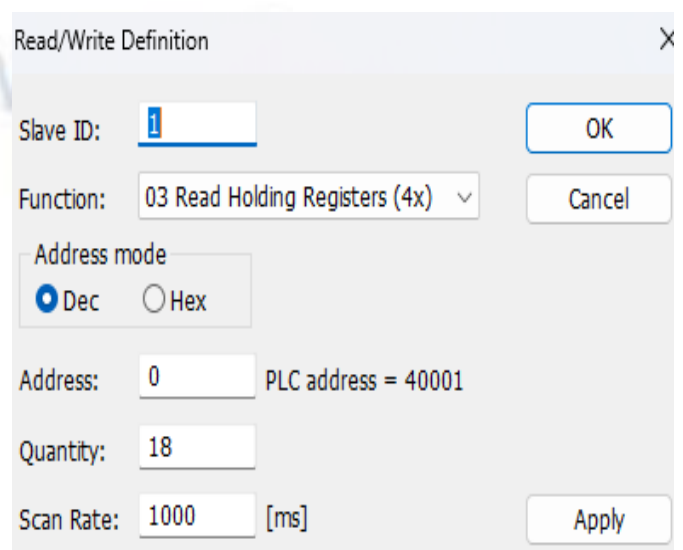
Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502

Connect Timeout: 1000 [ms]

☒ IPv4 ☐ IPv6



Read/Write Definition

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms]

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	Temperature	320	Current Baud	960
1	Humidity	480	New Baud	0
2		0	Current SID	1
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

- Above image Shows sensor data over Modbus with default Modbus settings
- At the register 40001 shows Temperature Values, register 40002 shows Humidity Values.
- The value 320 indicates 32.0 temperature and 480 indicates as 48.0 humidity.
- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	Temperature	320	Current Baud	960
1	Humidity	480	New Baud	11520
2		0	Current SID	1
3		0	New SID	10
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

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4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

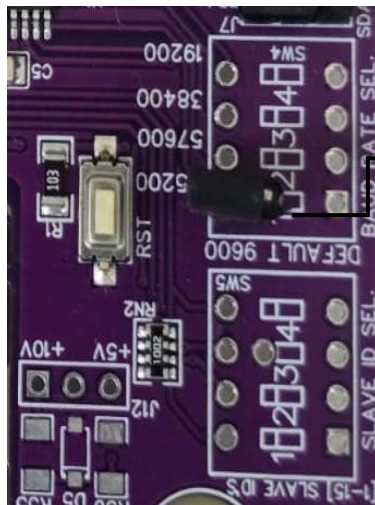


Image1.1



Image1.2

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

Contact Information

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